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OJT Optimizer: A Data-Driven Approach to OJT Selection

CCS 224 - Algorithms and Complexity

Final Project

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Project Overview

Introduction to the Project

On-the-Job Training (OJT) is one of the most important developmental opportunities for students transitioning from university to the professional world. However, selecting the right OJT placement is often a difficult and overwhelming process. Students must balance multiple factors such as allowance, company reputation, skills match, location, and personal career goals, all while facing time pressure and limited guidance.

This project, OJT Optimizer, aims to assist students in making smarter and faster OJT decisions. By allowing users to input and prioritize their preferences and by processing these through mathematical algorithms, the system generates a ranked list of OJT placements based on a calculated overall fit.

The main goal of the project is to provide a tool that eliminates guesswork and bias from the decision-making process, offering a data-driven, personalized recommendation to students. In doing so, this project hopes to reduce the stress and uncertainty that often accompany OJT placement and help students maximize their early career opportunities.

Problem Statement

Choosing the right internship is a critical step for college students preparing for real-world careers. However, many students struggle to objectively compare internship offers that vary across different factors such as distance, allowance, skills match, or company reputation, often leading to choices based on incomplete information or personal bias. Internships or on-the-job training (OJT) are mandatory for graduation under several academic programs, as stated in the Commission on Higher Education (CHED) Memorandum Order No. 104, Series of 2017. This requirement makes it even more important for students to carefully select internship placements that align with their career goals and personal priorities. Without a structured, objective method to evaluate multiple internship offers, students risk choosing suboptimal opportunities that may not maximize their skills, experience, or future employment potential. This project addresses this problem by using mathematical algorithms to systematically rank internship options based on user-defined criteria, helping students make more informed and confident decisions.

Significance in Algorithms and Complexity

The OJT Optimizer is significant in the field of algorithms and complexity because it demonstrates how mathematical methods can be used to solve real-world decision-making problems that involve multiple factors and constraints. Rather than relying on subjective or manual selection processes, this system applies linear algebra algorithms to compute optimal rankings efficiently and objectively.

Specifically, the system will use algorithms like Gaussian Elimination to combine weighted factors to calculate a score for each internship, QR Decomposition for normalising factors to prevent bias from one factor (like salary) dominating others unfairly, and Cramer's Rule to solve ranking priorities if the user sets very specific “must-have” constraints (like "must be remote"). These algorithms not only reduce the time complexity compared to brute-force or manual ranking but also ensure that the results are mathematically consistent and scalable even if the number of internships grows.

Algorithm Selection

After thorough research, these three algorithms were selected for this project: ***Gaussian Elimination, QR Decomposition, and Cramer's Rule***.

1. **Gaussian Elimination**

Gaussian Elimination is classified as a system of linear equations solver. It operates by applying a series of row operations to convert the original system into an upper triangular form, where all elements below the main diagonal are zero. Once the system is in this form, it can be solved easily through back-substitution, starting from the last equation and working upward. In the context of this project, Gaussian Elimination plays a critical role in solving the system of weighted criteria scores assigned to each internship option, thereby producing the initial ranked list that reflects the student's stated preferences. In the OJT Optimizer, Gaussian Elimination is used primarily during the initial ranking process. When a user inputs their prioritized criteria and internship options, the system constructs a weighted matrix representing the relationships between the internships and the criteria. Gaussian Elimination then solves this system to determine the best overall ranking based on the calculated scores. This method ensures that the final ranking is mathematically consistent with the user's preferences.

By applying Gaussian Elimination, it can systematically compute the rankings of internships based on the users' preferences, ensuring the results are accurate and consistent without the need for manual calculation.

2. QR Decomposition

QR Decomposition is a matrix factorization algorithm that decomposes a given matrix into two matrices: an orthogonal matrix (Q) and an upper triangular matrix (R). This decomposition is beneficial when solving linear systems, especially when numerical stability and orthogonality are important. In this project, QR Decomposition is vital during the preprocessing stage. Since internship evaluation factors such as allowance, distance, and reputation may have vastly different units or ranges, QR Decomposition allows data normalization so that all factors contribute fairly to the final ranking without being disproportionately influenced by their magnitude.

QR Decomposition is used to preprocess the input matrices. Since different criteria may be measured in different units (for instance, allowance in pesos and distance in kilometers), direct comparison without normalization would lead to biased results. QR Decomposition allows standardization of these different scales so that no single criterion unfairly dominates the scoring process. This preprocessing step is crucial in maintaining the fairness and integrity of the final ranking output. QR Decomposition is widely used in areas like numerical analysis, data science, and machine learning, especially for solving least squares problems where an exact solution is hard to find. It helps standardize and normalize different types of data so they can be fairly compared. In the OJT Optimizer, QR Decomposition is important because the chosen criteria—such as company reputation or distance—may have different scales. QR Decomposition helps adjust these differences, allowing the system to fairly combine and compare all factors to generate a balanced and meaningful ranking for students.

3. Cramer's Rule

Cramer's Rule is another algorithm for solving systems of linear equations, but it takes a different approach by using determinants. Specifically, it solves for each variable by replacing one column of the coefficient matrix with the result vector and computing the ratio of two determinants. While Cramer's Rule is computationally expensive for very large systems, it is highly effective and quick for small systems or when only one part of the system changes. Therefore, in this project, Cramer's Rule is applied when users modify their input after the initial ranking generation. Rather than recomputing the entire system using Gaussian Elimination, which could be time-consuming, Cramer's Rule allows the system to quickly update the rankings by focusing only on the changed parts, thus offering a better user experience through faster response times. Cramer's Rule is incorporated to handle dynamic interactions after the initial ranking. If a student wishes to adjust their priorities, for example, increasing the weight of "skills match" over "allowance", or add a new internship option, Cramer's Rule provides a fast recalculation method. Instead of repeating the entire Gaussian Elimination process, which could slow down the system, Cramer's Rule updates only the necessary parts, resulting in real-time responsiveness without sacrificing accuracy. This makes this system not only accurate but also user-friendly and adaptable to changing preferences.

Algorithm Comparison

Strengths and Weaknesses

- **Gaussian Elimination:**
  - **Strengths:**
    - Efficiently solves systems of linear equations by systematically reducing them to an upper triangular form, enabling straightforward solutions through back-substitution.
    - Provides a structured and well-established method for finding solutions, widely applicable in various fields like engineering, physics, and economics.
    - Plays a critical role in the OJT Optimizer by determining the initial ranked list of internship options based on weighted criteria scores.
  - **Weaknesses:**
    - It can become computationally intensive for very large systems of equations, potentially increasing processing time.
- **QR Decomposition:**
  - **Strengths:**
    - Offers numerical stability and orthogonality, which are important for solving linear systems and ensuring accurate results.
    - Effectively normalizes data from different sources or scales, allowing for uniform processing and fair comparison of internship evaluation factors.
    - Vital in the preprocessing stage of the project to prevent bias from factors with larger magnitudes, such as allowance, from disproportionately influencing the final ranking.
  - **Weaknesses:**
    - Involves matrix factorization, which can be more complex to implement compared to Gaussian Elimination.
- **Cramer's Rule:**
  - **Strengths:**
    - Provides a quick and effective solution for small systems of linear equations.

    - It is highly efficient for updating solutions when only a part of the system changes, making it suitable for dynamic user interactions.
    - Applied in the project to handle user modifications after the initial ranking, such as adjusting priorities or adding new internship options, ensuring real-time responsiveness.
  - **Weaknesses:**
    - Becomes computationally expensive for very large systems, limiting its practicality in such cases.
    - Less efficient than Gaussian Elimination for solving large systems from scratch.

2. Criteria for Comparison

- **Efficiency:**
  - Gaussian Elimination is efficient for solving systems of equations, particularly in the initial ranking process, providing a structured approach to obtain solutions.
  - QR Decomposition is efficient in the preprocessing stage. It ensures fair comparisons by normalizing data with different units or scales.
  - Cramer's Rule offers efficiency for dynamic updates, allowing quick recalculations when user preferences or internship options change, enhancing user experience.
- **Complexity:**
  - Gaussian Elimination's computational complexity varies with the size of the system of equations.
  - QR Decomposition involves matrix factorization, adding to its computational complexity.
  - Cramer's Rule's complexity increases significantly with the size of the system due to the computation of determinants.
- **Suitability:**
  - Gaussian Elimination is best suited for the initial ranking of internships, effectively processing weighted criteria to generate a base ranking.
  - QR Decomposition is ideal for preprocessing input matrices, normalizing data to ensure fair weighting of different criteria like allowance and distance.
  - Cramer's Rule is most suitable for dynamically updating rankings, providing quick adjustments when users modify their preferences or add new internship options.

Why was LU Decomposition not chosen?

Both LU decomposition and Gaussian elimination are fundamental for solving linear equations, with Gaussian elimination effectively performing LU decomposition; the choice depends on application, and Gaussian elimination's directness may be preferred for initial solutions. QR decomposition, preferred for numerical stability and normalization via orthogonalization, is crucial for handling diverse criteria in the Internship Position Ranker, unlike LU decomposition, which focuses on solving linear systems. While Cramer's rule uses determinants to directly solve linear systems, it's inefficient for large matrices; LU decomposition is generally more efficient, but the project uses Cramer's rule for efficient ranking updates, where recalculating with LU decomposition would be less efficient.

Hardware Requirements

Processor: 1.0 GHz dual-core processor - The planned algorithm is light and does not require significant computational resources; therefore, a low-powered CPU is sufficient.

RAM: 8 GB RAM - is enough to store the program data and run the operating system without running out of memory for small tasks.

Disk Space: 10GB available disk space - Only minimal storage is needed to install the programming environment and save a few small program files.

Special Considerations:

- No special hardware required.
- Basic CPU is sufficient; no GPU or parallel processing needed.

Software Requirements

Software Tools and Resources

1. **Operating System**
    - **Windows**: The application can be developed and run on Windows, which is widely used and supports various development tools.
2. **Development Environment**
    - **Integrated Development Environments (IDEs)**:
        - **Visual Studio Code**: A lightweight, versatile code editor that supports multiple programming languages and various extensions for enhanced functionality.
3. **Version Control System**
    - **Git**: A version control system for tracking changes in the codebase, facilitating collaboration among team members.

Programming Languages and Libraries

1. **Programming Languages**
    - **Python**: Chosen for its readability and extensive libraries that simplify mathematical computations and data handling, making it ideal for implementing algorithms.
    - **Java**: An alternative option for those familiar with object-oriented programming, providing robust libraries for data manipulation and algorithm implementation.
2. **Libraries and Frameworks:**

- NumPy: For numerical computing with support for multi-dimensional arrays.
- SciPy: For optimization and linear algebra.
- Pandas: For data manipulation and analysis.
- Matplotlib: This is used to create visualizations of ranking results.
- Scikit-learn: For data mining and potential machine learning applications.